

☒ EASY

☐ ALGEBRA

Mixed

10 questions · ~15 min

01

Q1

ALGEBRA

$$F = 2.50x + 7.00y$$

In the equation above, F represents the total amount of money, in dollars, a food truck charges for x drinks and y salads. The price, in dollars, of each drink is the same, and the price, in dollars, of each salad is the same. Which of the following is the best interpretation for the number 7.00 in this context?

- A. The price, in dollars, of one drink
- B. The price, in dollars, of one salad
- C. The number of drinks bought during the day
- D. The number of salads bought during the day

Q2

ALGEBRA

The function h is defined by $hx = x + 200$. What is the value of $h50$?

- A. 200
- B. 250
- C. 10,000
- D. 50,200

Q3

ALGEBRA

Jay walks at a speed of 3 miles per hour and runs at a speed of 5 miles per hour. He walks for w hours and runs for r hours for a combined total of 14 miles. Which equation represents this situation?

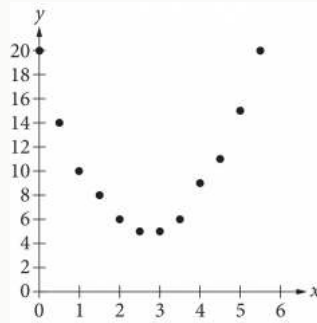
- A. $3w + 5r = 14$
- B. $\frac{1}{3}w + \frac{1}{5}r = 14$
- C. $\frac{1}{3}w + \frac{1}{5}r = 112$
- D. $3w + 5r = 112$

Q4

ALGEBRA

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters cm, of the fluid in the beaker x days after the start of the experiment. Which of the following is the best interpretation of 39 in this context?

- A. The estimated height, in cm, of the fluid at the start of the experiment
- B. The estimated height, in cm, of the fluid at the end of the experiment
- C. The estimated change in the height, in cm, of the fluid each day
- D. The estimated number of days for all the fluid to evaporate



Of the following, which is the best model for the data in the scatterplot?

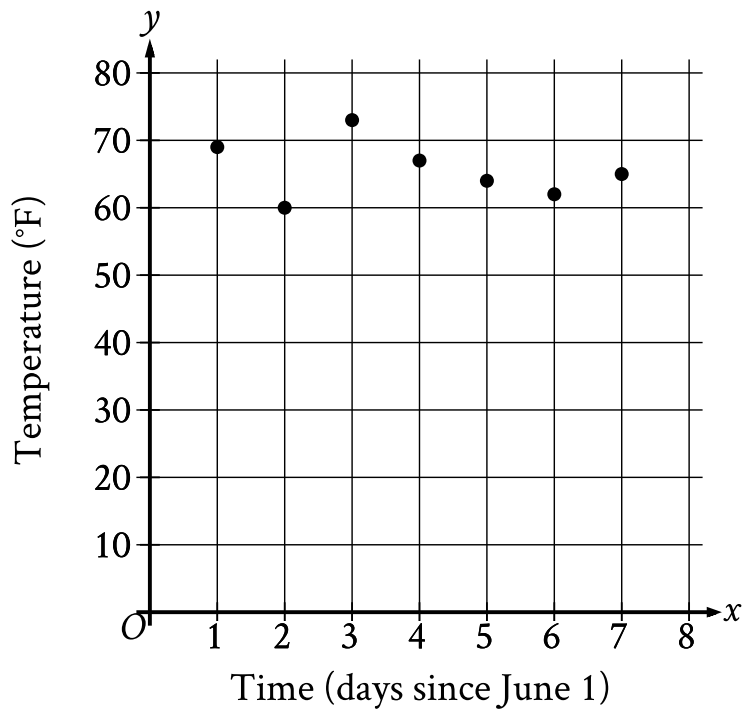
A. $y = 2x^2 - 11x - 20$

B. $y = 2x^2 - 11x + 20$

C. $y = 2x^2 - 5x - 3$

D. $y = 2x^2 - 5x + 3$

The scatterplot shows the temperature y , in $^{\circ}\text{F}$, recorded by a meteorologist at various times x , in days since June 1.

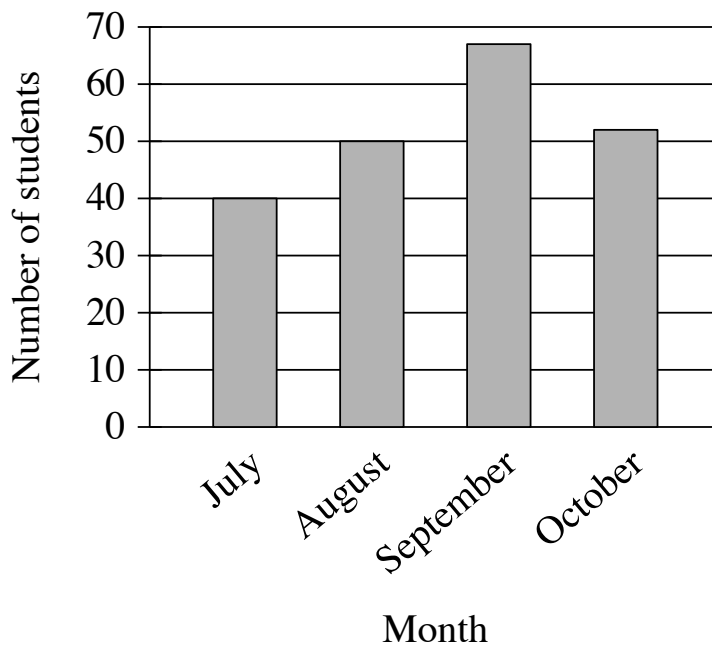


- The scatterplot has 7 data points.
- The data points are in a linear pattern trending approximately horizontally.
- The data points have the following coordinates:
 - (1 comma 69)
 - (2 comma 60)
 - (3 comma 73)
 - (4 comma 67)
 - (5 comma 64)
 - (6 comma 62)
 - (7 comma 65)

During which of the following time periods did the greatest increase in recorded temperature take place?

- A. From $x = 6$ to $x = 7$
- B. From $x = 5$ to $x = 6$
- C. From $x = 2$ to $x = 3$
- D. From $x = 1$ to $x = 2$

The bar graph shows the distribution of the number of students from one school who were born in one of four months.

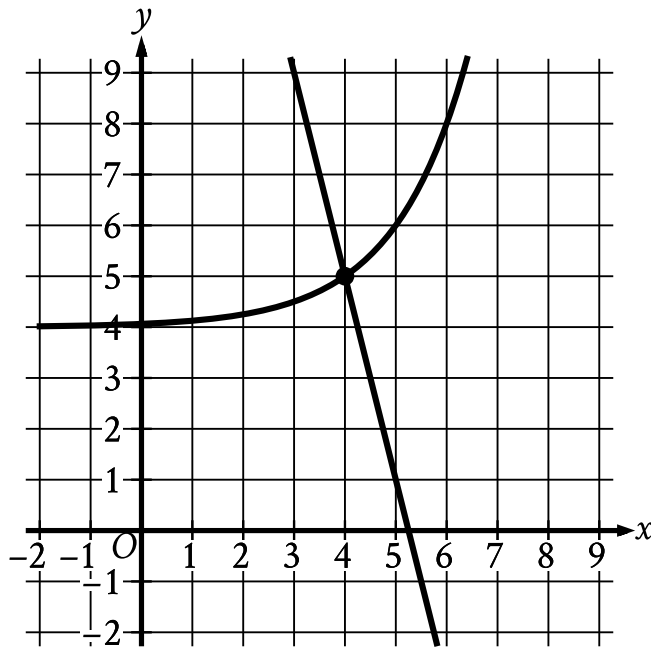


- The Number of students data for the 4 bars are as follows:
 - July: 40
 - August: 50
 - September: 67
 - October: 52

How many more students were born in August than were born in July?

- A. 90
- B. 50
- C. 40

D. 10



- For the curve in the system:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to the approximate point (0 comma 4.063).
 - In quadrant 1:
 - The curve trends up gradually to point (4 comma 5).
 - The curve then trends up sharply.
 - The curve passes through the following points:
 - (3 comma nine halves)
 - (4 comma 5)
 - (5 comma 6)
- For the line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (3 comma 9)
 - (4 comma 5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

- A. 0,0
- B. 0,4
- C. 4,5
- D. 5,0

Q9

ADVANCED MATH

Which expression is equivalent to $12x^3 - 5x^3$?

- A. $7x^6$
- B. $17x^3$
- C. $7x^3$
- D. $17x^6$

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY

Each side of a square has a length of 45. What is the perimeter of this square?

STUDENT-PRODUCED RESPONSE

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Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.